LAMP-Merge: Long-tail-Aware Medical Prototype Merging for One-shot Asynchronous Clinical Models

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Abstract

Model merging aims to integrate multiple independently trained local checkpoints into a unified global model, typically through general-purpose paradigms such as parameter interpolation, task-vector composition, or update-direction alignment. These methods implicitly assume that client models share relatively complete and comparable global discriminative structures. We argue that this assumption does not hold in medical settings, giving rise to two diagnostic-structure anomalies: (1) Aggregation-induced prediction collapse after merging: When clients have incomplete diagnostic-class coverage, multiple locally effective models can be merged into a globally collapsed model whose predictive distribution concentrates on a small number of dominant classes. (2) Long-tailed medical prevalence cannot be ignored. Majority classes in medical data often reflect genuine clinical prevalence rather than merely data noise. To address these challenges, we propose LAMP-Merge, which recasts medical model merging from global parameter interpolation as class-level diagnostic-evidence reconstruction. Through diagnostic prototype reconstruction(DPR) and long-tail prevalence calibration(LPC), LAMP-Merge constructs a global diagnostic model without exposing raw medical images or requiring multi-round federated communication. Experiments on the Blood, Derma, Organ-C, Organ-S, and Ultrasound medical imaging datasets show that LAMP-Merge outperforms existing model-merging methods in most client-averaged settings while substantially mitigating prediction collapse.

Introduction

Federated learning (McMahan et al. 2017; Li et al. 2020; Karimireddy et al. 2020; Kairouz et al. 2021) and model merging (Izmailov et al. 2018; Wortsman et al. 2022; Ilharco et al. 2023; Yadav et al. 2023) provide two important collaboration routes for privacy-sensitive multicenter medical intelligence. Federated learning enables hospitals to train collaboratively without sharing raw data (Sheller et al. 2020; Rieke et al. 2020), while model merging further reduces communication and optimization costs by integrating independently trained local models after training (Ainsworth, Hayase, and Srinivasa 2023; Stoica et al. 2024; Wu et al. 2025). These

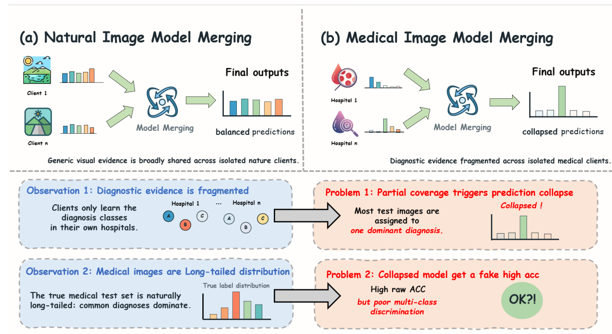


Figure 1: In multicenter healthcare settings, individual clients typically observe only a subset of the global diagnostic space. Conventional parameter-level model-merging methods can therefore cause the merged global model's predictions to concentrate on a small number of dominant classes.

paradigms are particularly attractive for medical imaging, because a single hospital rarely covers the full disease spectrum, while privacy regulations and device heterogeneity make centralizing raw medical images difficult (Kaissis et al. 2020a; Bonawitz et al. 2019; Abadi et al. 2016).

Despite their success on natural images and general visual tasks (Izmailov et al. 2018; Wortsman et al. 2022; Ilharco et al. 2023; Entezari et al. 2022; Ainsworth, Hayase, and Srinivasa 2023), transferring these paradigms to medical data remains challenging. A key reason is that general-purpose model merging implicitly assumes that different client checkpoints contain relatively complete and comparable global discriminative structures; local models are expected to align through parameter space, task-vector space, update signs, or curvature statistics (Yadav et al. 2023; Yu et al. 2024; Wu et al. 2025; Jin et al. 2025). This assumption does not hold in medical scenarios. Medical centers differ not only in device domains but also in diagnostic-class availability: some centers may mainly cover common diseases or specific examination types, while having few rare-disease samples (Guan and Liu

2022; Zech et al. 2018; Li et al. 2021). The resulting local models are not complete replicas of the global task, but partial diagnostic experts supported by locally observed diseases.

As illustrated in Fig. 1, we summarize these challenges as two diagnostic-structure abnormalities in medical model merging:

- Diagnostic evidence fragmented across isolated medical clients. Each client has reliable supervision only for observed classes; thus, a local model may be locally discriminative but still lack evidence for the full global diagnostic space (Guan and Liu 2022; Li et al. 2021).
- Long-tailed Medical Image Prevalence. Medical image datasets naturally follow a long-tailed distribution: a few common diagnostic classes dominate the samples, while rare classes are underrepresented (Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010). Under this imbalanced class prior, a merged model collapsed to the majority class can still obtain high Accuracy despite lacking complete multi-class discrimination.

To address these issues, we propose LAMP-Merge, a long-tail-aware medical prototype-merging method for single-shot asynchronous medical image model merging. LAMP-Merge shifts the merge target from global parameter interpolation to class-level diagnostic-evidence reconstruction. The key idea is to use client-uploaded class-level statistics as a synchronization signal for the global diagnostic space, allowing the server to reconstruct a global diagnostic classifier without accessing raw medical images, using a server-side validation set, or conducting multi-round federated communication. Specifically, LAMP-Merge consists of two mechanisms: diagnostic prototype reconstruction aggregates reliable class-level evidence through prototypes in a shared reference feature space, and long-tail prevalence calibration injects a bounded prior bias from client-uploaded class counts, thereby mitigating prediction collapse while preserving clinically meaningful majority-class prevalence.

Our main contributions are summarized as follows:

- Medical Merging Diagnosis. We revisit model merging under single-shot asynchronous medical collaboration and identify limited local diagnostic coverage and aggregation-induced prediction collapse as two key abnormalities caused by medical class availability and long-tailed prevalence.
- LAMP-Merge. We propose a long-tail-aware medical prototype-merging method that reconstructs a global diagnostic classifier from client-side class prototypes and prevalence statistics without exposing raw images or requiring multi-round federated optimization.
- Empirical Validation. We conduct full-scope experiments on five medical image datasets, four vision-model families, three client-population sizes, and three non-IID skew levels, showing that LAMP-Merge consistently outperforms strong model-merging baselines and improves prediction-collapse diagnostics.

Related Works

Model Merging

Model-merging methods integrate independently trained checkpoints in parameter, task-vector, or geometric spaces. Weight averaging (Izmailov et al. 2018) and Model Soups (Wortsman et al. 2022) directly interpolate parameters; permutation-aware matching (Entezari et al. 2022; Ainsworth, Hayase, and Srinivasa 2023) aligns hidden-unit symmetries; ZipIt (Stoica et al. 2024) performs feature-level zipping; Task Arithmetic (Ilharco et al. 2023), TIES-Merging (Yadav et al. 2023), and DARE (Yu et al. 2024) manipulate task deltas and sign conflicts; Fisher merging (Wu et al. 2025), RegMean (Jin et al. 2025), Model Stock (Jang, Yun, and Han 2025), Breadcrumbs (Davari and Belilovsky 2024), Iso-C (Marczak et al. 2025), FreeMerge (Zheng and Wang 2025), and RobustMerge (Zeng et al. 2025) introduce curvature, mean-statistic, subspace, frequency-domain, or robustness assumptions.

However, these methods share a class-agnostic merging interface: their variables are global parameters, task vectors, update signs, or geometric statistics rather than class-level diagnostic evidence. They do not encode whether a client has reliable supervision for a specific medical class. Under incomplete local class coverage and long-tailed prevalence, reliable class directions can be mixed with absent or conflicting directions from other clients, resulting in aggregation-induced prediction collapse(Buda, Maki, and Mazurowski 2018; Zech et al. 2018).

Medical Model Merging

Medical fusion studies exploit domain structures such as irregular clinical time series, physiological sensor graphs, and image-EHR fusion. mTAN (Shukla and Marlin 2021), Raindrop (Zhang et al. 2022), and MedFuse (Hayat, Geras, and Shamout 2023) show that medical tasks benefit from modeling clinical sampling mechanisms, modality heterogeneity, and patient-level temporal relations.

However, their fusion objects are usually patient-level inputs, multimodal records, or clinical representations rather than independently trained image-classifier checkpoints. They often require patient samples, timestamps, electronic health records, or server-side validation feedback, which conflicts with common privacy constraints (Kaissis et al. 2020a; Sheller et al. 2020; Bonawitz et al. 2019). MedMerge (Wei et al. 2026) is a relevant training-free medical knowledge-transfer framework, but it targets cross-modal adaptation for medical vision-language models rather than

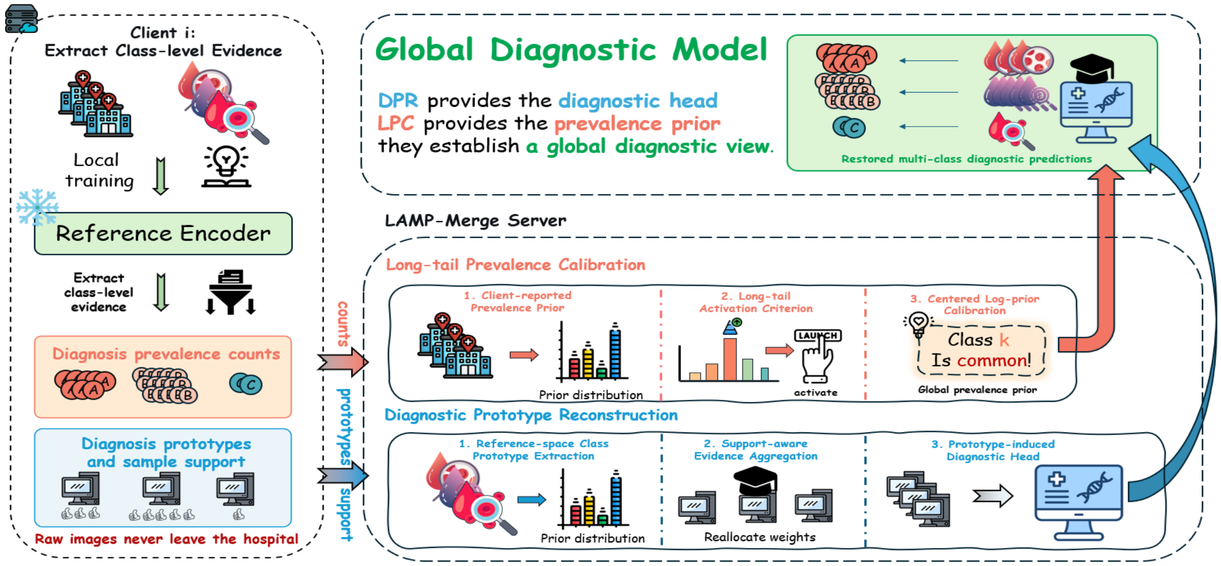


Figure 2: Local models from multicenter hospitals typically contain only partial diagnostic knowledge, while raw medical images cannot be centrally shared. LAMP-Merge reconstructs a privacy-friendly global medical classifier through DPR and LPC, without accessing raw data or requiring multi-round federated optimization.

asynchronous checkpoint merging for the same classification task. Thus, existing medical fusion routes are mismatched with our privacy and single-upload protocol constraints. LAMP-Merge keeps the medical structure uploadable by using only locally computed class prototypes and prevalence counts.

Federated Learning

Federated and decentralized learning perform collaborative training through communication protocols. FedAvg (McMahan et al. 2017) and its variants or medical deployments (Li et al. 2020; Karimireddy et al. 2020; Li et al. 2021; Kaissis et al. 2020b; Sheller et al. 2020; Rieke et al. 2020; Kairouz et al. 2021) rely on repeated server-client updates; Swarm Learning (Warnat-Herresthal et al. 2021) and Gossip Learning (Heged\Hus, Danner, and Jelasity 2019) reduce centralization but still require participants to continue training, exchanging models, and optimizing from intermediate states.

However, our setting begins after local training has finished. Hospitals may not remain online, receive global checkpoints repeatedly, or run additional optimization rounds. The bottleneck is therefore communication feasibility in addition to privacy. LAMP-Merge requires one upload of a local model interface and class-level statistics, enabling the server to construct a unified diagnostic model without a multi-round feedback loop.

Method

In this section, we first formulate the diagnostic-structure pathologies exposed when generic model merging is applied to medical data, namely limited local diagnostic coverage and aggregation-induced prediction collapse. To correct these intrinsic mismatches, we propose LAMP-Merge, a two-stage framework composed of DPR and LPC, as shown in Fig. 2.

Preliminaries

Problem Formulation. Assume K medical clients. Client i owns a private labeled dataset $D_i = \{(x, y)\}$ drawn from a class-skewed medical distribution over C diagnosis classes. Let $D_{i,c}$ denote the subset of client i belonging to diagnosis class c. In medical partitions, $D_{i,c}$ can be empty for many classes; therefore, client i has reliable supervision only for its observed classes. The goal is to construct a global diagnostic model under this partial and heterogeneous diagnostic-label coverage.

Upload and Server Constraints. We study single-shot asynchronous merging: after local training, each client uploads only once, while the server cannot access raw medical images, per-sample features, per-sample predictions, or a server-side validation set, and does not issue multi-round updates to clients. Clients are allowed to upload class-level statistics. Specifically, $n_{i,c}$ denotes the class-support count, $m_{i,c}$ denotes the class-prevalence count, and $\mu_{i,c}$ denotes the class prototype computed in a shared reference feature space. The server performs one merge using only the local checkpoint interfaces and these class-level statistics. This setting differs from federated learning because clients do not continue optimization from server feedback; it also differs from standard model merging because the presence of class-level diagnostic evidence becomes an explicit merging variable.

Motivation

Although generic model merging is training-free and parameter-efficient, we find that its diagnostic structure can severely degenerate in medical model merging. Prediction-distribution diagnosis and class-support analysis reveal two key abnormalities:

(1) Aggregation-induced prediction collapse. Under partial diagnostic coverage, parameter-level merging can turn several locally useful predictors into a globally collapsed predictor. The merged model tends to allocate most test images to a few dominant classes, indicating that global class-discriminative structure has not been reconstructed.

(2) Prevalence prior cannot be ignored. Medical datasets often exhibit clinically meaningful long-tailed prevalence. Therefore, completely suppressing majority-class dominance is also inappropriate: a collapsed predictor is undesirable, but a merger should still preserve calibrated disease prevalence when the dominant class reflects the true clinical prior.

Diagnostic Prototype Reconstruction Module

To mitigate the fragmented diagnostic evidence across isolated medical clients caused by heterogeneous diagnostic coverage and conflicting class evidence, the diagnostic prototype reconstruction module shifts the merging target from parameter averaging to class-conditional evidence reconstruction. Instead of directly averaging local classifiers, it first collects reliable evidence for each diagnosis class in a shared reference space and then synthesizes a global diagnostic head over the full label space.

(1) Reference-space Class Prototype Extraction. First, all clients extract class prototypes with the same frozen reference backbone. Let ϕ_0 be the reference backbone instantiated from the shared model configuration before local training, and let $T(\cdot)$ be the evaluation transform. For each diagnosis class c, client i locally computes a reference-space class prototype

$$\mu_{i,c} = \frac{1}{n_{i,c}} \sum_{(x,y)\in D_i} \mathbf{1}[y = c]\phi_0(T(x)), \tag{1}$$

where $n_{i,c} = |D_{i,c}|$ is the class support count. If $n_{i,c} = 0$, the client uploads zero support for that class and contributes no prototype evidence.

(2) Support-aware Evidence Aggregation. Second, the server converts class-support counts into evidence weights to characterize the reliability of each client for a given class:

$$e_{i,c} = (n_{i,c} + 1)^{\gamma}\mathbf{1}[n_{i,c} > 0], \tag{2}$$

where $\gamma = 0.55$ is the default evidence exponent. This weight gives larger class-specific contribution to clients with more supervised samples of class c, while excluding clients that never observe this class. The normalized reliability is

$$\alpha_{i,c} = \frac{e_{i,c}}{\sum_{j=1}^{K} e_{j,c}}. \tag{3}$$

The global diagnosis prototype is obtained by class-wise reliability-weighted aggregation:

$$p_c = \sum_{i=1}^{K} \alpha_{i,c}\mu_{i,c}. \tag{4}$$

This step aggregates evidence only from clients that have actually observed class c, preventing absent-class classifiers from diluting valid diagnostic signals.

(3) Prototype-induced Diagnostic Head. Finally, LAMP-Merge maps global diagnosis prototypes into a cosine prototype classifier:

$$w_c = s\frac{p_c}{\|p_c\|_2}, \tag{5}$$

where $s = 18.75$ is the default head scale. The resulting head is not an average of local classifier parameters; it is induced by class evidence in the shared reference space, thereby restoring multiclass diagnostic directions over the global label space.

Long-tail Prevalence Calibration

To address the fact that prevalence prior cannot be ignored in real medical tasks, the long-tail prevalence calibration module introduces a bounded class prior after prototype reconstruction. This module does not push the model back to single-class collapse; instead, it calibrates the clinical prevalence reflected by client statistics after multiclass diagnostic directions have been restored.

(1) Client-reported Prevalence Prior. First, each client uploads class-prevalence counts $m_{i,c}$. The server estimates the global class prior

$$\pi_c = \frac{\sum_{i=1}^{K} m_{i,c}}{\sum_{k=1}^{C}\sum_{i=1}^{K} m_{i,k}}. \tag{6}$$

This prior is computed only from class-level counts, not from raw images, per-sample features, or per-sample predictions.

(2) Long-tail Activation Criterion. Second, to avoid unnecessary calibration in non-long-tailed cases, the server measures dominant-class imbalance by

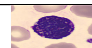
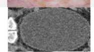
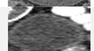
$$r = C \max_c \pi_c. \tag{7}$$

The calibration module is activated only when $r > \tau$, where $\tau = 2.5$ by default.

(3) Centered Log-prior Calibration. Finally, when the module is activated, LAMP-Merge constructs a centered log-prior bias

$$b_c = \lambda\left(\log \pi_c - \frac{1}{C}\sum_{k=1}^{C} \log \pi_k\right), \tag{8}$$

Table 1: Dataset statistics and representative samples for the five medical image benchmarks. Counts are taken from the exact NPZ files used in our experiments.

Dataset	Classes	Train	Val	Test	Total	Sample
Blood	8	11,959	1,712	3,421	17,092	
Derma	7	7,007	1,003	2,005	10,015	
Organ-C	11	12,975	2,392	8,216	23,583	
Organ-S	11	13,932	2,452	8,827	25,211	
Ultrasound	8	3,869	549	1,113	5,531	

with default maximum calibration strength $\lambda = 4.25$. The centering term prevents a uniform shift of all class logits, so the bias only changes the relative class prior. This module calibrates prevalence without replacing the diagnostic directions reconstructed by the first module.

Optimization Objective and Algorithm

The final scoring function of LAMP-Merge integrates the diagnostic prototype reconstruction term and the long-tail prevalence calibration term:

$$\ell_c(x) = w_c^\top \phi_0(T(x)) + \mathbf{1}[r > \tau]\, b_c. \tag{9}$$

Experiments

Experimental Settings

(1) Evaluation Datasets: We evaluate the merged models on five representative MedMNIST v2 benchmarks(Yang, Shi, and Ni 2023): Blood, Derma, Organ-C, Organ-S, and Ultrasound. They cover blood-cell microscopy, dermoscopy, abdominal CT, and ultrasound image classification (detailed dataset descriptions are provided in the Appendix).

(2) Baselines: We compare LAMP-Merge with twelve model-merging baselines covering the main technical routes of current model merging: avg, ties(Yadav et al. 2023), dare_linear(Yu et al. 2024), dare_ties(Yu et al. 2024), regmean(Jin et al. 2025), fisher(Wu et al. 2025), breadcrumbs(Davari and Belilovsky 2024), model_stock(Jang, Yun, and Han 2025), from(Ilharco et al. 2023), iso_c(Marczak et al. 2025), free_merge(Zheng and Wang 2025), and robust-merge(Zeng et al. 2025). These baselines include weight averaging, task-vector pruning, sign-conflict resolution, curvature or mean-statistic correction, stock-style model interpolation, isotropic subspace merging, Fourier-domain fusion, and direction-robust merging.

(3) Merging Settings: We evaluate four vision-model families: ResNet, ConvNeXt, ViT-Tiny, and Swin-Tiny(He et al. 2016; Liu et al. 2022; Dosovitskiy et al. 2021; Liu et al. 2021). For each dataset-model pair, we construct non-IID client partitions with $K \in \{3, 5, 7\}$ clients and Dirichlet skew parameter $\beta \in \{0, 0.01, 0.1\}$, using seed 42(Hsu, Qi, and Brown 2019). A client-average result fixes the dataset, model, and K, and then averages over the three β values. And the experiments follow a single-shot asynchronous model-merging protocol.

Comparison With the State of the Arts

Tables 2–5 report the full client-average results for all four backbones. Across the 60 client-average cells, LAMP-Merge wins or ties the strongest non-LAMP baseline in 57 cells (95.0%).

Table 2: Client-average test accuracy (%) on ResNet. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg
Weight Avg.	33.45	24.67	28.22	28.78	62.91	47.00	62.44	57.45	22.61	17.72	13.96	18.10	28.64	17.94	24.75	23.78	24.35	15.81	15.42	18.53
TIES[NeurIPS 2023]	26.01	18.72	17.87	20.87	59.95	48.41	64.84	57.73	26.50	16.84	21.18	21.51	19.81	19.15	18.74	19.23	34.77	19.89	18.54	24.40
DARE-Linear[ICML 2024]	28.32	28.56	23.92	26.93	63.57	48.41	62.88	58.29	20.18	25.88	16.50	20.85	25.45	22.45	25.37	24.42	21.62	17.28	16.08	18.33
DARE-TIES[ICML 2024]	31.94	19.35	17.80	23.03	53.75	48.25	55.58	52.53	18.51	15.15	18.59	17.42	17.90	13.52	17.30	16.24	29.32	16.35	18.90	21.52
RegMean[ICLR 2023]	34.60	23.69	34.07	30.79	58.12	28.43	50.22	45.59	24.05	13.40	13.88	17.11	25.04	16.23	14.93	18.73	22.76	16.95	13.18	17.63
Fisher[NeurIPS 2022]	31.96	22.37	26.08	26.80	67.07	67.02	66.90	67.00	29.73	17.41	18.12	21.75	20.10	29.56	19.63	23.10	23.33	18.60	25.13	22.35
Breadcrumbs[ECCV 2024]	8.54	14.17	14.60	12.44	4.95	29.13	17.21	17.10	7.66	6.76	6.85	7.09	8.02	9.48	6.03	7.84	12.25	12.73	10.21	11.73
Model Stock[ECCV 2024]	24.20	18.24	19.75	20.73	62.16	43.11	57.21	54.16	12.25	12.74	11.85	12.28	27.42	15.78	20.12	21.11	17.91	13.21	15.78	15.63
Iso-C[ICML 2025]	32.78	27.48	30.23	30.16	50.49	44.22	64.16	52.96	23.31	16.29	13.83	17.81	27.36	21.18	24.53	24.36	26.15	16.98	15.54	19.56
Free-Merging[ICCV 2025]	33.45	24.67	28.22	28.78	62.91	47.00	62.46	57.46	22.61	17.71	13.96	18.09	28.63	17.93	24.75	23.77	24.35	15.81	15.45	18.54
RobustMerge[NeurIPS 2025]	34.01	24.30	22.56	26.96	29.56	20.85	39.42	29.94	24.58	17.64	16.53	19.58	26.80	22.27	23.01	24.03	21.89	22.31	17.07	20.42
FROM[AAAI 2026]	29.89	23.06	20.52	24.49	67.13	43.69	65.80	58.87	23.50	15.60	15.20	18.10	22.07	20.87	24.76	22.57	23.09	15.06	20.63	19.59
LAMP-Merge	80.01	79.94	79.92	79.95	67.55	67.63	67.63	67.60	60.05	59.94	60.07	60.02	52.98	52.79	52.81	52.86	48.07	46.75	47.35	47.39

Table 3: Client-average test accuracy (%) on ConvNeXt. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg
Weight Avg.	15.65	11.22	13.84	13.57	29.66	48.30	66.88	48.28	10.37	12.18	9.44	10.66	13.08	10.35	14.16	12.53	15.06	15.06	12.52	14.21
TIES[NeurIPS 2023]	17.15	12.14	15.24	14.84	29.66	17.97	27.13	24.92	12.17	7.86	7.16	9.06	11.23	7.61	7.96	8.93	15.18	14.56	12.91	14.22
DARE-Linear[ICML 2024]	17.56	11.22	13.45	14.08	27.05	48.30	48.30	41.22	10.35	6.99	10.85	9.40	12.82	10.35	7.97	10.38	15.75	15.75	12.64	14.71
DARE-TIES[ICML 2024]	10.96	15.19	14.97	13.71	9.08	27.98	24.51	20.52	13.05	13.46	7.70	11.40	10.96	9.46	7.09	9.17	14.56	12.97	15.00	14.18
RegMean[ICLR 2023]	15.65	8.17	18.56	14.13	26.33	7.73	27.20	20.42	11.43	10.17	7.68	9.76	10.14	12.59	10.38	11.04	15.75	15.39	15.00	15.38
Fisher[NeurIPS 2022]	15.65	9.99	11.89	12.51	48.25	26.38	29.71	34.78	10.03	9.74	9.15	9.64	6.04	5.17	8.43	6.55	16.95	13.60	14.56	15.04
Breadcrumbs[ECCV 2024]	15.16	13.66	11.63	13.48	5.24	27.66	46.30	26.40	10.31	14.44	15.74	13.50	15.16	15.54	20.77	17.16	13.84	13.63	13.63	13.70
Model Stock[ECCV 2024]	17.56	11.44	17.78	15.59	29.66	48.30	66.88	48.28	12.76	12.18	13.85	12.93	13.08	10.35	19.35	14.26	15.75	12.91	14.79	14.48
Iso-C[ICML 2025]	19.47	18.65	14.80	17.64	29.71	48.30	66.88	48.30	9.91	6.99	9.45	8.78	7.89	14.77	8.67	10.44	15.18	12.91	16.05	14.71
Free-Merging[ICCV 2025]	15.75	11.90	13.84	13.83	29.71	48.30	66.88	48.30	9.06	12.18	13.85	11.70	13.08	10.35	14.16	12.53	15.06	12.91	16.95	14.97
RobustMerge[NeurIPS 2025]	14.10	14.05	7.52	11.89	9.08	45.07	48.30	34.15	13.94	12.18	17.93	14.68	23.54	9.61	20.77	17.97	13.00	11.35	13.63	12.66
FROM[AAAI 2026]	15.65	10.83	16.71	14.40	29.66	34.06	45.69	36.47	12.32	12.24	6.50	10.35	13.24	11.05	8.97	11.09	15.75	15.78	10.87	14.13
LAMP-Merge	84.85	84.87	84.98	84.90	59.68	58.77	59.42	59.29	65.41	65.33	65.51	65.42	60.55	60.50	60.66	60.57	43.07	42.80	42.41	42.76

Table 4: Client-average test accuracy (%) on ViT-Tiny. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg
Weight Avg.	13.43	10.82	11.60	11.95	46.38	43.69	36.96	42.34	16.57	10.43	9.52	12.17	8.44	8.52	17.10	11.35	14.50	13.27	18.84	15.54
TIES[NeurIPS 2023]	7.96	14.21	12.85	11.67	47.80	34.98	46.15	42.98	13.41	12.23	9.53	11.72	11.92	10.02	15.97	12.64	20.01	13.06	10.18	14.42
DARE-Linear[ICML 2024]	23.97	11.29	13.87	16.38	28.50	10.02	12.04	16.85	15.34	9.46	7.04	10.61	7.48	5.49	16.11	9.69	13.66	10.51	14.35	12.84
DARE-TIES[ICML 2024]	7.35	11.89	11.46	10.23	27.35	45.02	35.28	35.88	12.52	9.03	10.33	10.63	5.41	8.74	5.91	6.69	14.47	11.20	10.42	12.03
RegMean[ICLR 2023]	17.48	14.17	12.90	14.85	51.14	36.67	66.58	51.46	12.50	9.64	9.83	10.66	12.63	7.17	10.68	10.16	14.91	13.84	11.44	13.40
Fisher[NeurIPS 2022]	11.66	15.12	19.44	15.41	47.80	20.42	48.20	38.81	7.99	10.72	14.24	10.98	8.56	12.18	12.22	10.99	16.11	13.15	14.20	14.49
Breadcrumbs[ECCV 2024]	13.63	10.39	13.00	12.34	4.89	25.72	6.07	12.23	7.73	10.00	16.44	11.39	12.37	4.83	12.19	9.80	13.60	11.41	15.24	13.42
Model Stock[ECCV 2024]	19.40	11.20	11.34	13.98	38.05	30.79	27.35	32.06	14.88	12.27	12.15	13.10	8.93	8.59	18.67	12.06	15.99	18.09	16.41	16.83
Iso-C[ICML 2025]	12.34	14.83	10.18	12.45	57.09	45.29	23.06	41.81	10.84	5.67	8.97	8.49	6.11	5.31	10.81	7.41	14.17	15.27	18.69	16.04
Free-Merging[ICCV 2025]	19.89	14.57	13.92	16.13	46.30	47.70	38.55	44.18	14.25	9.80	7.59	10.55	7.27	9.14	16.43	10.95	14.05	15.93	16.47	15.48
RobustMerge[NeurIPS 2025]	14.82	10.70	14.25	13.26	3.47	27.68	17.24	16.13	9.54	6.17	14.38	10.03	12.60	7.07	12.17	10.61	16.98	18.54	15.63	17.05
FROM[AAAI 2026]	18.02	7.81	14.05	13.29	46.95	51.21	66.88	55.01	21.48	13.01	9.72	14.74	7.72	7.98	14.85	10.18	18.36	17.43	17.76	17.85
LAMP-Merge	85.21	85.06	84.98	85.09	59.45	59.32	59.83	59.53	58.78	58.40	58.29	58.49	54.39	54.51	54.65	54.52	46.12	45.97	45.94	46.01

Table 5: Client-average test accuracy (%) on Swin-Tiny. Each entry averages the three Dirichlet settings for a fixed client number; Avg denotes the mean over $K = 3, 5, 7$ within the corresponding dataset.

Method	Blood				Derma				Organ-C				Organ-S				Ultrasound			
	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg	$K=3$	$K=5$	$K=7$	Avg
Weight Avg.	15.65	11.86	15.24	14.25	45.69	48.30	66.88	53.62	10.68	12.94	7.87	10.50	6.82	6.04	13.81	8.89	15.27	12.94	14.65	14.29
TIES[NeurIPS 2023]	9.72	11.90	13.84	11.82	8.41	7.80	66.88	27.70	11.60	7.02	9.34	9.32	7.34	6.07	11.07	8.16	16.95	13.03	12.61	14.20
DARE-Linear[ICML 2024]	15.65	11.61	18.25	15.17	45.69	27.71	46.30	39.90	11.55	12.72	7.76	10.68	6.40	7.20	13.81	9.14	17.43	15.36	14.97	15.92
DARE-TIES[ICML 2024]	13.84	11.62	13.84	13.10	29.61	26.38	66.88	40.96	12.76	7.74	11.16	10.55	11.94	4.34	12.30	9.53	17.91	13.51	13.00	14.81
RegMean[ICLR 2023]	13.74	17.15	15.78	15.56	27.05	46.30	45.69	39.68	14.11	7.73	8.11	9.98	12.15	8.21	9.26	9.87	14.65	11.05	17.79	14.50
Fisher[NeurIPS 2022]	16.33	17.56	15.65	16.51	45.82	6.52	47.18	33.17	8.14	8.99	9.78	8.97	6.51	6.50	10.92	7.98	17.46	11.65	17.55	15.55
Breadcrumbs[ECCV 2024]	15.00	13.65	15.37	14.67	6.47	27.71	7.15	13.78	14.45	17.83	16.60	16.29	23.54	20.77	21.11	21.81	14.50	13.12	16.53	14.72
Model Stock[ECCV 2024]	13.74	14.40	17.15	15.10	45.69	27.71	48.25	40.55	11.10	7.71	14.04	10.95	6.71	11.06	17.14	11.64	15.39	12.73	18.69	15.60
Iso-C[ICML 2025]	14.30	17.22	17.00	16.17	30.99	28.40	51.74	37.04	13.02	13.27	14.69	13.66	15.45	17.27	18.80	17.17	17.55	17.31	17.91	17.59
Free-Merging[ICCV 2025]	11.54	8.61	18.40	12.85	45.69	48.30	66.88	53.62	9.59	13.77	8.90	10.75	6.82	4.97	10.62	7.47	15.18	15.75	15.99	15.64
RobustMerge[NeurIPS 2025]	13.74	14.09	16.70	14.84	23.77	27.71	5.04	18.84	16.01	14.42	19.60	16.68	16.73	11.51	20.77	16.34	17.61	13.15	16.29	15.68
FROM[AAAI 2026]	15.65	14.95	16.71	15.77	45.69	29.71	66.88	47.43	11.74	12.20	10.51	11.48	11.33	10.16	13.80	11.76	15.39	12.52	14.76	14.22
LAMP-Merge	77.38	77.34	77.17	77.30	63.24	63.18	63.09	63.17	68.49	68.17	68.12	68.26	62.58	62.80	62.84	62.74	47.71	48.70	48.40	48.27

Table 6: DPR internal ablation by dataset. Each entry averages all backbones, client counts, and Dirichlet settings; Avg averages the five datasets.

Category	Setting	Blood	Derma	Organ-C	Organ-S	US	Avg
Prototype semantics	CHA	18.95	56.14	17.74	18.88	16.43	25.63
	GFM	8.88	66.88	22.33	23.54	10.87	26.50
	SSH	11.62	17.29	7.08	7.74	10.69	10.89
	SLP	24.46	28.73	14.18	16.25	14.25	19.57
Client-support weighting	UCW	78.03	59.61	58.36	53.86	41.79	58.33
	BSO	78.03	44.96	57.86	53.24	41.79	55.18
	GCSW	78.45	62.48	59.11	53.79	43.13	59.39
	LAMP	81.81	62.40	63.05	57.67	46.11	62.21

Table 7: LPC internal ablation by dataset. UPP uses a uniform prior, AOC removes the activation gate, and CBP averages client-normalized priors. Each entry averages all backbones, client counts, and Dirichlet settings; Avg averages the five datasets.

Setting	Blood	Derma	Organ-C	Organ-S	US	Avg
UPP	81.81	46.58	62.57	57.50	46.11	58.91
AOC	80.57	62.40	63.05	57.67	45.80	61.90
CBP	81.25	49.61	62.10	57.12	44.76	58.97
LAMP-Merge	81.81	62.40	63.05	57.67	46.11	62.21

Ablation Study

Module-level ablations use exactly the same experimental setup as the main experiments. The Weight-Averaging Baseline, Baseline + DPR, and full LAMP-Merge achieve overall client-average ACC values of 22.04%, 58.91%, and 62.21%, respectively. We additionally measure the complete DPR/LPC factorial effect through the strict 2×2 TIES/DARE controls in Fig. 4.

1) Effectiveness of Diagnostic Prototype Reconstruction To further identify which information within diagnostic prototype reconstruction is necessary, we conduct controlled substitutions across the full setting. All configurations retain long-tail prevalence calibration and vary only the prototype semantics or support-count weighting.

Table 6 groups the DPR controls into prototype semantics (CHA, GFM, SSH, and SLP) and client-support weighting (UCW, BSO, and GCSW). Neither an arbitrary head nor a class-agnostic statistic explains the gain; replacing reference-space prototypes or class-specific support weights consistently reduces ACC, confirming that diagnostic semantics and class-level reliability are jointly necessary.

2) Effectiveness of Long-tail Prevalence Calibration We further conduct three full-scope LPC internal controls while retaining DPR. Uniform prevalence prior (UPP) sets $\pi_c = 1/C$, removing empirical long-tail prevalence. Always-on calibration (AOC) replaces the

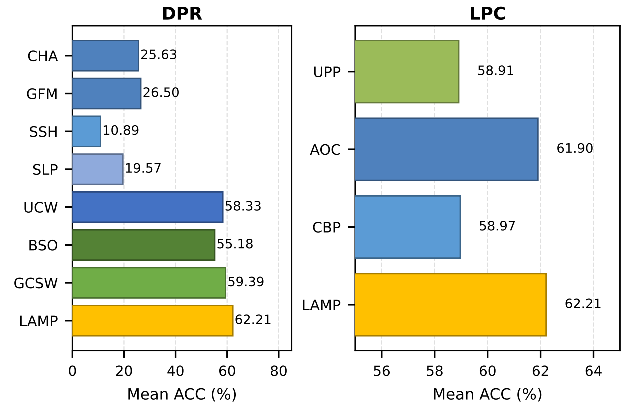


Figure 3: Internal ablations of the Diagnostic Prototype Reconstruction (DPR) and Long-tail Prevalence Calibration (LPC) modules. Full names and configurations of the plotted abbreviations are provided in the supplementary ablation configurations.

thresholded activation $\mathbf{1}[r > \tau]$ with 1 while retaining the empirical prior. Client-balanced prior (CBP) first normalizes each client's prevalence counts and then averages clients, rather than weighting the global prior by raw client counts. Table 7 reports overall mean ACC values of 58.91% (UPP), 61.90% (AOC), and 58.97% (CBP), compared with 62.21% for LAMP-Merge. Thus, UPP, AOC, and CBP are lower by 3.30, 0.31, and 3.24 percentage points, respectively. The small AOC gap indicates that always applying the empirical prior is close but still inferior, whereas UPP and CBP show that both the empirical long-tail prior and its client-size-aware aggregation are needed. The strict controls in Fig. 4 cross DPR and LPC for each classifier-head baseline. Without DPR, LPC raises TIES from 19.52% to 28.21% and DARE from 19.33% to 28.79%. DPR raises the two baselines to 58.91%, and adding LPC reaches 62.21%. Thus, DPR provides the dominant improvement, while LPC supplies a complementary gain in the complete method.

Prediction-collapse Diagnostics

Although the main experiments use ACC for direct comparison, long-tailed medical distributions can allow single-class or few-class predictors to achieve high ACC(Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010). We therefore report collapse ratio, effective predicted classes, and predicted–true total variation.

Table 8 compares LAMP-Merge with the best-performing baseline for each metric. LAMP-Merge consistently and substantially outperforms the existing methods.

Visualization Experiment

To visualize the prediction-distribution changes measured by the diagnostic metrics, we plot the absolute

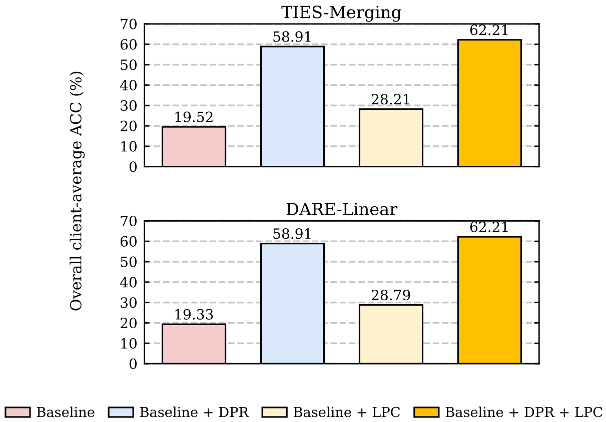


Figure 4: Strict 2×2 DPR/LPC ablations for TIES-Merging and DARE-Linear. All experimental datasets are included.

Table 8: Prediction-collapse diagnostics by dataset. Best ref. is selected separately for each metric and dataset. Ratio metrics use percentage scale; Avg averages the five datasets.

Metric	Series	Blood	Derma	Organ-C	Organ-S	US	Avg
Collapse ↓	Best ref.	80.20	85.18	72.38	71.12	80.31	79.02
	LAMP	19.58	68.05	20.72	24.36	20.45	30.63
Eff. classes ↑	Best ref.	1.81	1.65	2.49	2.53	1.98	2.05
	LAMP	7.48	3.10	9.55	8.44	7.44	7.20
Pred-true TV ↓	Best ref.	73.48	46.03	75.42	73.85	72.73	69.44
	LAMP	3.97	17.09	12.38	13.17	15.51	12.42

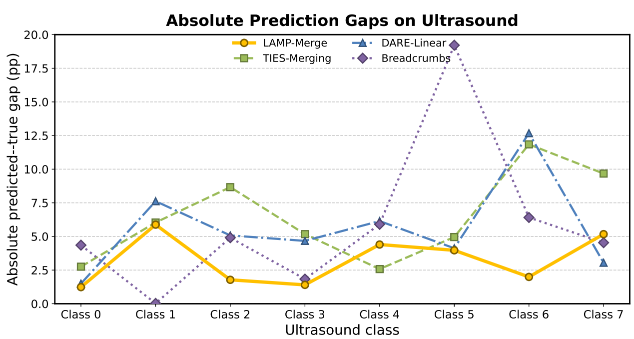


Figure 5: Absolute per-class prediction gaps on Ultrasound, averaged over the four visual backbones, $K \in \{3, 5, 7\}$, and the three Dirichlet skew levels. Values are absolute predicted–true class-proportion differences in percentage points; zero denotes exact distributional agreement.

per-class gap between each method's predicted allocation and the true test distribution. Values closer to zero indicate better prevalence alignment. LAMP-Merge exhibits smaller deviations than representative generic merges across most classes.

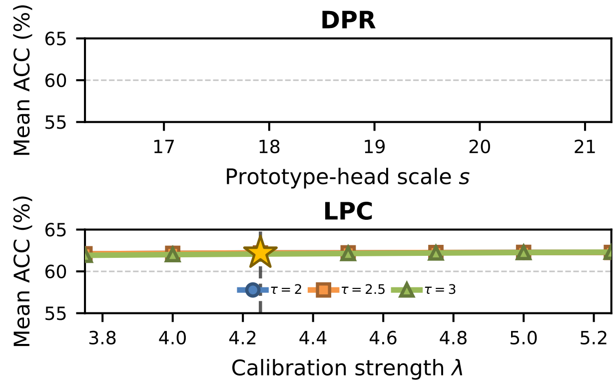


Figure 6: DPR and LPC sensitivity under the full LAMP-Merge configuration. Completed curve points average 60 client-average cells.

Hyperparameter Analysis Experiment

Fig. 6 evaluates LPC sensitivity under the full LAMP-Merge configuration. Each curve point uses 180 raw cells and 60 client-average cells. The sweep fixes DPR at $\gamma = 0.55$ and $s = 18.75$ while varying τ and λ around the selected operating point.

Conclusion

We studied one-shot asynchronous medical model merging, where independently trained hospital models must be fused without federated optimization rounds and without exposing raw medical images to the server. Our empirical analysis shows that generic model-merging methods suffer from aggregation-induced predictive collapse under incomplete local diagnostic coverage. We further show that, on long-tailed medical datasets, such collapse can be masked by raw accuracy. We therefore introduce the Long-tail Prevalence Calibration module and evaluate prediction-collapse diagnostics alongside accuracy(Buda, Maki, and Mazurowski 2018; Brodersen et al. 2010).

Taken together, the full-scope comparisons, module ablations, and prediction-distribution diagnostics establish a consistent mechanism: class-level support-weighted evidence reconstructs discriminative directions for locally absent diagnoses, while bounded prevalence calibration retains clinically meaningful imbalance without reintroducing majority-class collapse.

LAMP-Merge addresses these two properties through a concise two-module design. Future work will extend its applicability in two directions: first, from medical image classification to segmentation, detection, and multimodal diagnostic tasks; and second, toward stronger privacy-preserving mechanisms, such as differential privacy and secure aggregation, to further reduce potential privacy risks associated with class-level statistics(Kaissis et al. 2020a; Geyer, Klein, and Nabi 2017; Abadi et al. 2016). We will also explore deployment and validation in larger-scale real-world multicenter healthcare settings.

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Supplementary Results and Analysis

Evaluation Datasets

The five MedMNIST v2 benchmarks(Yang, Shi, and Ni 2023) cover blood-cell microscopy, dermoscopy, abdominal CT organ slices, and ultrasound images. They involve cell morphology recognition, skin-lesion diagnosis, organ-structure recognition, and ultrasound-structure discrimination:

- Blood: an eight-class blood-cell microscopy dataset for hematological morphology classification, focusing on fine-grained cellular appearance and inter-class morphological similarity.
- Derma: a seven-class dermoscopic skin-lesion dataset, characterized by strong long-tailed prevalence and visually subtle lesion categories.
- Organ-C: an 11-class abdominal CT dataset in the coronal view, evaluating organ-category recognition under grayscale cross-sectional imaging.
- Organ-S: an 11-class abdominal CT dataset in the sagittal view, complementing Organ-C with a different anatomical imaging plane.
- Ultrasound: an eight-class ultrasound image dataset, covering structure recognition under modality-specific speckle noise and low-contrast boundaries.

Together, these benchmarks provide a systematic testbed for evaluating cross-modality robustness, class-discriminative capability, and task generalization under different medical image types and class-distribution conditions.

Detailed Ablation Configurations

All diagnostic-prototype substitutions retain long-tail prevalence calibration and vary only prototype semantics or support-count weighting:

- LAMP-Merge uses $p_c = \sum_i \alpha_{i,c}\mu_{i,c}$, where $e_{i,c} = (n_{i,c}+1)^{\gamma}\mathbf{1}[n_{i,c} > 0]$ determines the class-support reliability.
- Classifier-head aggregation (CHA) replaces the reference-space class prototype by the local classifier-head direction, i.e., $\mu_{i,c} \leftarrow h_{i,c}$.

- Global-feature mean (GFM) removes class conditioning by setting $\mu_{i,c} \leftarrow g$ for all c, where g is the class-agnostic global reference-feature mean.
- Support-only synthetic head (SSH) removes image evidence by replacing each prototype with a fixed random unit direction u_c, i.e., $p_c \leftarrow u_c$, $\|u_c\|_2 = 1$.
- Shuffled-label prototype (SLP) preserves prototype magnitudes but permutes diagnostic semantics, i.e., $p_c \leftarrow p_{\sigma(c)}$, where σ is a random permutation over diagnosis labels.
- Uniform client weight (UCW) replaces support-weighted aggregation with equal weights among clients containing class c, i.e., $\alpha_{i,c} \leftarrow \mathbf{1}[n_{i,c} > 0]/\sum_j \mathbf{1}[n_{j,c} > 0]$.
- Binary support only (BSO) replaces fine-grained support and prevalence counts with class-presence indicators, i.e., $n_{i,c}, m_{i,c} \leftarrow \mathbf{1}[n_{i,c} > 0]$.
- Global client-size weight (GCSW) uses the total client support $N_i = \sum_k n_{i,k}$ rather than class-specific support, i.e., $e_{i,c} \leftarrow (N_i + 1)^\gamma \mathbf{1}[n_{i,c} > 0]$.

The long-tail prevalence controls retain DPR and vary only the prior estimator or activation gate:

- LAMP-Merge estimates the empirical class prior as $\pi_c = \sum_i m_{i,c}/\sum_k \sum_i m_{i,k}$.
- Uniform prevalence prior (UPP) removes long-tail prevalence by setting $\pi_c \leftarrow 1/C$, where C is the number of diagnosis classes.
- Always-on calibration (AOC) removes the long-tail activation gate by replacing $\mathbf{1}[r > \tau]$ with 1, while retaining the empirical prior and centered log-prior bias.
- Client-balanced prior (CBP) gives each client equal influence by setting $\pi_c \leftarrow K^{-1} \sum_i m_{i,c}/(\sum_k m_{i,k})$, while retaining the activation gate and centered log-prior bias.

2) Additional Theoretical Analysis Let $\mu_c^\star$ denote the true class mean in the shared reference space. If client class features have bounded second moments, the prototype estimation error satisfies

$$\mathrm{E}\|p_c - \mu_c^\star\|_2^2 \le \sum_i \alpha_{i,c}^2 \frac{\sigma_c^2}{n_{i,c}} + \mathrm{Bias}_c^2. \tag{10}$$

This upper bound shows that support counts directly control the variance of class-prototype estimation. A client that does not observe class c should not contribute evidence for that class, and a client with more samples should contribute more, but only sublinearly.

Let the true margin between classes c and d for sample x be

$$\Delta_{c,d}(x) = (\mu_c^\star)^\top z - (\mu_d^\star)^\top z. \tag{11}$$

When the prototype error satisfies

$$\|p_c - \mu_c^\star\|_2 + \|p_d - \mu_d^\star\|_2 < \frac{\Delta_{c,d}(x)}{\|z\|_2}, \tag{12}$$

the estimated prototypes preserve the class order between c and d.

The role of long-tail prevalence calibration is to introduce long-tail priors in a bounded manner. For prevalence counts $m_{i,c}$, the class prior is

$$\pi_c = \frac{\sum_i m_{i,c}}{\sum_k \sum_i m_{i,k}}, \tag{13}$$

and the centered log-prior bias is

$$b_c = \lambda \left(\log \pi_c - \frac{1}{C} \sum_k \log \pi_k \right). \tag{14}$$

As long as λ is bounded, long-tail prevalence calibration does not replace the discriminative directions reconstructed by diagnostic prototype reconstruction; it only calibrates the output toward the real class prevalence. This explains why using long-tail prevalence calibration alone is insufficient, while full LAMP-Merge remains both non-collapsed and prevalence-aware.

Metric Definitions

Let y_n and $\hat{y}_n$ denote the true and predicted labels of the n-th test sample. Let $q_c = N^{-1} \sum_n \mathbf{1}[\hat{y}_n = c]$ be the predicted-class distribution and $p_c = N^{-1} \sum_n \mathbf{1}[y_n = c]$ the true class distribution. We use:

- $\rho = \max_c q_c$, the prediction-collapse ratio, which measures concentration on a single diagnosis class.
- $C_{\text{eff}} = \exp(-\sum_c q_c \log q_c)$, the effective number of predicted classes.
- $\mathrm{TV}(q, p) = \frac{1}{2} \sum_c |q_c - p_c|$, the total variation distance between predicted and true class distributions.

Hyperparameter-sweep Reproducibility

The sensitivity figure uses the completed full-configuration LPC sweep only. It uses $\tau \in \{2.0, 2.5, 3.0\}$ and ten calibration strengths $\lambda \in \{3.00, 3.25, \ldots, 5.25\}$, while holding $\gamma = 0.55$ and $s = 18.75$ fixed. At every grid point, we evaluate the five datasets, four backbone families, three client counts, and three Dirichlet skew levels, yielding 180 raw result cells and 60 client-average cells after averaging the three skew levels for each dataset–backbone–client-count combination.

For reuse, the paper package records one row per grid point in the DPR and LPC hyperparameter CSV files. Each row includes both the raw-cell and client-average-cell counts together with the mean ACC used in Fig. 6. The curves retain every measured point in ascending parameter order and do not apply smoothing, interpolation, or best-run selection. The marked operating point is fixed before comparison at $\gamma = 0.55$, $s = 18.75$, $\tau = 2.5$, and $\lambda = 4.25$; it is therefore directly traceable to the same recorded grid as the surrounding sensitivity values.